STAGE 1: DATA PREPARATION

Raw data → one clean weekly table

- One clean weekly table:
  5 counties, 26 years (2000–2025)
- Weather rolled into 1–30-day sums:
  local + upstream, rivers & lakes
- Flood history & time-of-year,
  encoded as predictive features
- No peeking ahead: only data
  known by forecast Friday

STAGE 2: HAZARD ENGINE

Level-1 · County Probabilistic Engine

- Multi-Head LightGBM
  (50 weekly engineered features)
- Predicts flood occurrence probability
  & multi-horizon duration (1–4 wks)
- Quantile area regression: q10 (best),
  q50 (median), q90 (worst)

STAGE 3: PIXEL DOWNSCALER

Level-2 · Spatial Allocation (~232m)

- 26-year conditional pixel climatology
  (MODIS MCDWD)
- Golden Rule: allocates WHERE water
  pools based on county q50
- Coordinate snapping guarantees exact
  area conservation (0.0 km² error)

STAGE 4: ASSET RISK OVERLAP

Level-3 · Humanitarian Early Action

- Direct spatial raster multiplication
- ASAP Cropland Mask (~500m grid) →
  inundated staple hectares
- ICPAC Cattle Density (~9km grid) →
  exposed head of livestock